Helix Thready — Workable-Items Backlog (Full-Granularity Cards)

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Rev	Date	Author	Change
1	2026-07-22	swarm (development, pass 3)	Full-granularity expansion: per-item cards (phase → sub-phase → task) with Given/When/Then acceptance, dependencies, blocks, test-type coverage and verified-source anchors for ATM-001…ATM-072

This document is the **implementation-ready** expansion of the summary backlog in workable-items.md. The summary tables give the *shape* of the backlog (id, title, kind, priority, depends-on, gap); **this** document gives each item its full card: the decoupled scope, the **task-level sub-items** (`ATM-NNN.k`) a worker/subagent claims, the **Given/When/Then acceptance criteria**, the explicit **blocks/blocked-by** edges, the mandated **test types**, and — where a claim was verified against real module source during Pass 3 — a **Verified-source** anchor citing the exact file. Nothing here is a bluff: an item that depends on a scaffold/stub says so and names the hardening prerequisite.

The card format is the unit the orchestration model dispatches (see agent-orchestration.md §4–§6): a **track worker** claims an `ATM-NNN`, decomposes it into its `ATM-NNN.k` tasks, and dispatches a **subagent per task** on disjoint file globs. The `ATM-NNN.k` ids are the disjoint-scope guarantee `[§11.4.58 L3]` — two subagents never share a task's glob.

> **Provenance tags** `[CONSTITUTION §x]` · `[IN-HOUSE: module]` · `[RESEARCH]` · `[OPERATOR]` · `[DEFAULT — adjustable]` · `[BUILD-NEW]` · `[GAP: id]` · `[VERIFIED-SOURCE]` (read at module source during Pass 3) · `[OPEN: …]`.

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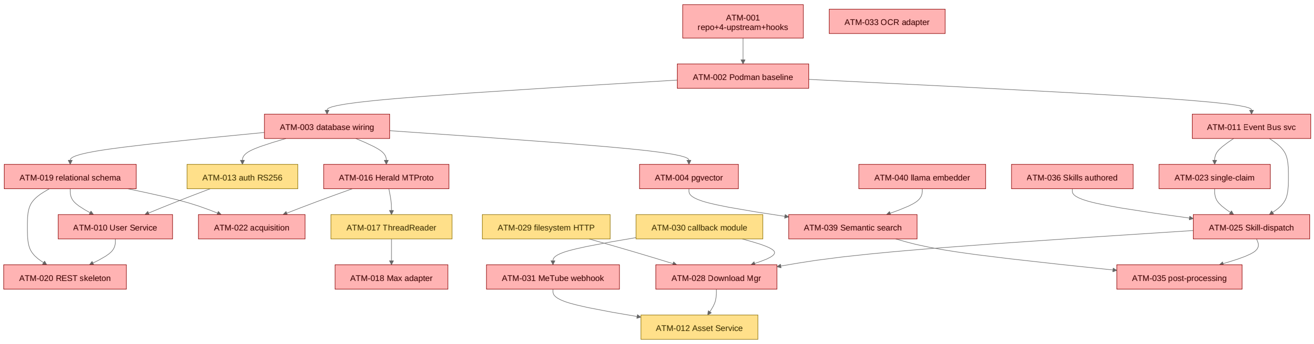
1. Card schema

Each card carries the fields below. Fields also present in the summary DB (`docs/workable_items.db`, `[§11.4.93/95]`) are mirrored 1:1; the **Tasks** and **Given/When/Then** fields are the new granularity this document adds.

- **Kind / Pri / Phase** — `PRODUCTION-WIRE | EXTEND | BUILD-NEW | HARDEN | AUDIT`; `P0|P1|P2`; `phase.subphase` from final request §5.1.2.
- **Provides / Depends on / Blocks** — the item's output contract and its explicit graph edges.
- **Decoupled scope** — what the item does and, crucially, what it must **not** touch `[§11.4.28]`.
- **Tasks (`ATM-NNN.k`)** — the disjoint sub-scopes a subagent claims; each names its file-glob home.
- **Acceptance (Given/When/Then)** — executable acceptance statements; these become the RED tests.
- **Test types** — the mandated subset of the 15 `[§11.4.27]`; mocks are unit-only.
- **Verified-source** — where a fact was read at module source in Pass 3 (else the caveat is kept).

2. Critical-path dependency DAG



Explanation (for readers/models that cannot see the diagram). This is the P0/P1 critical path the multi-track ruler follows; red is P0 (blocks the MVP) and amber is P1 (GA-grade). The single root is `ATM-001` (repo + four-upstream bootstrap + git hooks): nothing can be committed to spec until the enforcement layer exists, so it gates the entire tree. From the container baseline (`ATM-002`) the graph forks into three broad rivers that run **concurrently** on separate tracks: the *data river* (`ATM-003` database → `ATM-004` pgvector and `ATM-019` relational schema → `ATM-020` REST and `ATM-010` User Service), the *messenger river* (`ATM-016` Herald MTProto → `ATM-017` ThreadReader → `ATM-018` Max, and → `ATM-022` acquisition), and the *processing river* (`ATM-011` Event Bus → `ATM-023` single-claim + `ATM-025` Skill-dispatch).

The three rivers rejoin at the processing spine: the Skill-dispatch engine (`ATM-025`) drives the Download Manager (`ATM-028`), which — with the callback module (`ATM-030`) and the `filesystem` HTTP-source fix (`ATM-029`) — feeds the Asset Service (`ATM-012`) alongside the MeTube webhook (`ATM-031`). A parallel semantic branch runs `ATM-040` (enforce the real llama embedder) → `ATM-039` (Semantic search) fed by `ATM-004` (pgvector), and finally post-processing (`ATM-035`) consumes both the dispatch output and the semantic index to write the status reply and grow the Skill-Trees. The OCR adapter (`ATM-033`) hangs off the graph with no inbound edge because it is genuinely independent — it can proceed the moment a track is free. The ruler's job is to keep every one of these rivers saturated: a freed track is immediately re-assigned the next-highest-priority item whose inbound edges are all `DONE` `[§11.4.192]`.

> Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: diagrams/critical-path-dag.mmd.

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3. Phase 1 — Foundation (ATM-001…ATM-021)

3.1 Infrastructure (sub-phase 1.1)

ATM-001 — Repo + 4-upstream bootstrap + git hooks

- **Kind/Pri/Phase:** PRODUCTION-WIRE · P0 · 1.1. **Depends on:** — . **Blocks:** everything.
- **Provides:** a repo whose every commit is spec-enforced and fans out to all four upstreams.

- **Decoupled scope:** install the org tooling into `helix_thready`; do **not** author product code.
- **Tasks:**
  - `ATM-001.1` — place `upstreams/{GitHub,GitLab,GitFlic,GitVerse}.sh` (VERIFIED present) and run `install_upstreams.sh` so `github/gitlab/gitflic/gitverse` remotes exist. Glob: `upstreams/**`.
  - `ATM-001.2` — vendor `scripts/git_hooks/{pre-commit,pre-push,post-commit,commit-msg}` and run the **symlink** installer `scripts/install_git_hooks.sh`. Glob: `scripts/**`.
  - `ATM-001.3` — vendor `scripts/commit_all.sh` + `scripts/push_all.sh`; add `docs/audit/bypass_events.md`.
  - `ATM-001.4` — wire the private submodule `docs/private` (VERIFIED `.gitmodules`).
- **Acceptance (Given/When/Then):**
  - Given a fresh clone, When `install_upstreams.sh` runs, Then `git remote` lists exactly `github, gitlab, gitflic, gitverse` pointing at the SSH URLs in Appendix C.
  - Given a staged `.md` lacking `.html/.pdf` siblings in the governed set, When `git commit`, Then the `pre-commit` hook exits 1 and blocks.
  - Given `git commit --no-verify` with no `Bypass-rationale:` footer, When committing, Then the `commit-msg` hook blocks (marker-freshness detection).
- **Test types:** unit (hook logic), integration (real four-remote push to test repos), challenges.
- **Verified-source `[VERIFIED-SOURCE]`:** installer is symlink-based/idempotent (`helix_code/scripts/install_git_hooks.sh`); commit-msg uses the `.git/ATMO_PRECOMMIT_RAN` marker (`helix_code/scripts/git_hooks/commit-msg`); see git-workflow-internals.md.
- **Gap:** — . **Provenance:** `[CONSTITUTION §11.4.75/§2.1]` `[VERIFIED-SOURCE]`.

ATM-002 — Rootless Podman Compose baseline

- **Kind/Pri/Phase:** PRODUCTION-WIRE · P0 · 1.1. **Depends on:** ATM-001. **Blocks:** ATM-003/005/006/007/011/057.
- **Provides:** the sole orchestration substrate `[§11.4.76]` — `pkg/boot` / `pkg/compose` / `pkg/health`.
- **Decoupled scope:** stand up `vasic-digital/containers`; no product services yet.
- **Tasks:** `ATM-002.1` compose baseline + rootless Podman socket; `ATM-002.2` per-service health probes via `observability/pkg/health`; `ATM-002.3` three env profiles (`dev.` / `sta.` / `thready.`).
- **Acceptance:** Given the compose baseline, When `pkg/boot` starts, Then all declared containers report healthy within the boot budget and Podman runs **rootless** (no root daemon). Given a killed container, When health polling runs, Then it is reported unhealthy and restarted.
- **Test types:** unit, integration (real Podman), chaos (container kill), performance (boot time).
- **Gap:** — . **Provenance:** `[CONSTITUTION §11.4.76/161]` `[IN-HOUSE: containers]`.

ATM-003 — `digital.vasic.database` wiring

- **Kind/Pri/Phase:** PRODUCTION-WIRE · P0 · 1.1. **Depends on:** ATM-002. **Blocks:** ATM-004/013/019/043.
- **Provides:** SQLite dev (`modernc.org/sqlite` , cgo-free) / Postgres prod (`pgx/v5`) + `migration.Runner` .
- **Tasks:** `ATM-003.1` DSN/config injection (project-not-aware); `ATM-003.2` `migration.Runner` up/down harness; `ATM-003.3` connection-pool + health.
- **Acceptance:** Given the dev profile, When the app boots, Then it opens a cgo-free SQLite DB and applies all pending migrations; Given the prod profile, Then it opens Postgres via `pgx/v5` . Given a migration `up` then `down` , Then the schema round-trips with no residue (expand-contract, `[Q30]`).
- **Test types:** unit, integration (both backends), migration round-trip.
- **Gap:** — . **Provenance:** `[IN-HOUSE: database]` `[Q12/Q30]` .

ATM-004 — pgvector provisioning + `digital.vasic.vectordb`

- **Kind/Pri/Phase:** PRODUCTION-WIRE · P0 · 1.1. **Depends on:** ATM-003. **Blocks:** ATM-039/042.
- **Provides:** cosine `<=>` vector store co-located in Postgres behind the `VectorStore` seam.
- **Tasks:** `ATM-004.1` enable pgvector extension in the Postgres image; `ATM-004.2` `VectorStore` cosine upsert/search; `ATM-004.3` id-hydration join back to relational rows; `ATM-004.4` the embedding-provider guard (`RequireSemanticEmbedder` , ties to ATM-040).
- **Acceptance:** Given a set of embedded rows, When a cosine `<=>` top-k query runs, Then the returned ids hydrate against relational rows and the search completes < 500 ms (Aggressive SLO, `[Q14]`). Given `HELIX_EMBEDDING_PROVIDER != llama` in a RAG/search context, Then the guard fails loudly (never the `HashEmbedder` stub — see ATM-040).
- **Test types:** unit, integration, performance (< 500 ms), benchmarking.
- **Gap:** 3.1. **Provenance:** `[IN-HOUSE: vectordb]` `[GAP 2.1]` .

ATM-005 — `digital.vasic.observability`

- **Kind/Pri/Phase:** PRODUCTION-WIRE · P0 · 1.1. **Depends on:** ATM-002. **Blocks:** all runtime items (telemetry).
- **Provides:** OTel traces + Prometheus metrics + logrus structured logs + ClickHouse + `pkg/health` .
- **Tasks:** `ATM-005.1` OTLP exporter + Jaeger/Zipkin; `ATM-005.2` Prometheus registry + Grafana dashboards; `ATM-005.3` logrus + correlation-id middleware; `ATM-005.4` ClickHouse analytics sink.
- **Acceptance:** Given a cross-service call, When it executes, Then an OTel span with a correlation id is emitted and queue/latency/retry Prometheus metrics update. Given a secret in a log field, Then it is redacted (`[§11.4.10]`).
- **Test types:** unit, integration (real collectors), security (no-secret-in-log).
- **Gap:** — . **Provenance:** `[IN-HOUSE: observability]` `[Q40/Q42/Q43]` .

ATM-006 — Let's Encrypt TLS per subdomain

- **Kind/Pri/Phase:** PRODUCTION-WIRE · P1 · 1.1. **Depends on:** ATM-002. **Blocks:** ATM-057.
- **Provides:** ACME certs (HTTP-01/DNS-01) per `dev.` / `sta.` / `thready.` with atomic deploy-hook + rollback.
- **Tasks:** `ATM-006.1` acme.sh integration; `ATM-006.2` atomic deploy-hook; `ATM-006.3` systemd renew timer; `ATM-006.4` rollback-on-failed-reload.
- **Acceptance:** Given a subdomain, When issuance runs, Then a valid cert is installed atomically and a failed reload rolls back to the prior cert; When renewal is due, Then the timer renews without downtime.
- **Test types:** unit, integration (staging ACME), chaos (failed reload → rollback).
- **Gap:** — . **Provenance:** `[IN-HOUSE: lets_encrypt]` `[Q44]`.

ATM-007 — Dynamic ports (`discovery` + `mdns` + `port_prefix`)

- **Kind/Pri/Phase:** PRODUCTION-WIRE · P1 · 1.1. **Depends on:** ATM-002.
- **Provides:** deterministic dynamic port allocation ≤ 65535 + service registration + failover.
- **Tasks:** `ATM-007.1` `port_prefix` deterministic allocation; `ATM-007.2` `discovery` registration; `ATM-007.3` `mdns` advertisement; `ATM-007.4` health-gated deregistration.
- **Acceptance:** Given N services, When they boot, Then each gets a deterministic port ≤ 65535 with no collision and registers; When one dies, Then it deregisters and failover routes elsewhere.
- **Test types:** unit, integration, chaos (service death → failover).
- **Gap:** — . **Provenance:** `[IN-HOUSE: discovery/mdns/port_prefix]` `[§14.3]`.

ATM-008 — CodeGraph index of own-org submodules

- **Kind/Pri/Phase:** PRODUCTION-WIRE · P1 · 1.1. **Depends on:** ATM-001.
- **Provides:** a local SQLite code-intelligence index `[§11.4.78-80]` + scheduled re-index.
- **Tasks:** `ATM-008.1` index the reused submodules; `ATM-008.2` MCP wiring for agent queries; `ATM-008.3` scheduled re-index hook.
- **Acceptance:** Given the reused submodules, When indexing runs, Then `codegraph explore` answers symbol/call-path queries; When a submodule changes, Then the scheduled re-index refreshes it.
- **Test types:** unit, integration.
- **Gap:** — . **Provenance:** `[CONSTITUTION §11.4.78-80]` `[IN-HOUSE: codegraph]`.

ATM-009 — Docs Chain contexts (md → HTML/PDF/DOCX siblings)

- **Kind/Pri/Phase:** PRODUCTION-WIRE · P1 · 1.1. **Depends on:** ATM-001. **Related:** ATM-069.
- **Provides:** `.docs_chain/contexts/*.yaml` so every `.md` gets HTML/PDF/DOCX siblings `[§11.4.65]`.
- **Tasks:** `ATM-009.1` author the Thready docs-chain context; `ATM-009.2` wire `sync_all_markdown_exports.sh`; `ATM-009.3` provision pandoc/weasyprint (ties to ATM-069).

- **Acceptance:** Given a committed `.md`, When Docs Chain runs, Then HTML/PDF/DOCX siblings exist and the content-hash DAG marks them fresh; Given pandoc/weasyprint absent, Then it **honest-SKIPs** with a logged reason (never fakes success, `[§11.4.3/52]`).
- **Test types:** unit, integration (real pandoc/weasyprint), challenges (SKIP honesty).
- **Gap:** 10.1. **Provenance:** `[CONSTITUTION §11.4.65/106]` `[GAP 10.1]`.

3.2 Core Services (sub-phase 1.2)

ATM-010 — User Service `[BUILD-NEW]`

- **Kind/Pri/Phase:** BUILD-NEW · P0 · 1.2. **Depends on:** ATM-003, ATM-013. **Blocks:** ATM-012, ATM-020.
- **Provides:** three-tier multi-tenant RBAC (root / account-admin / user) + cross-account membership.
- **Decoupled scope:** own repo on `auth` + `security/pkg/policy` + Catalogizer RBAC pattern; **not** a vendored copy inside `helix_thready`. Design: build-new-subsystems.md §7.
- **Tasks:** `ATM-010.1` `BootstrapRoot` (owner-only, once, at deploy); `ATM-010.2` `CreateAccount` (Root-only); `ATM-010.3` `Invite` (Root or that account's Admin); `ATM-010.4` `Authorize` → `security/pkg/policy`; `ATM-010.5` TOTP MFA for admin tiers; `ATM-010.6` account/user/membership DDL.
- **Acceptance:** Given a fresh deploy, When bootstrap runs, Then exactly one Root Admin exists, owner-only. Given a non-Root caller, When `CreateAccount`, Then `403`. Given a user in two accounts, Then both memberships resolve and each permission check flows through `security/pkg/policy`. Given an admin login without TOTP, Then `401`.
- **Test types:** unit, integration, e2e, **security** (authz matrix + privilege-escalation + MFA bypass + token revocation), challenges, HelixQA.
- **Gap:** 7.2, §11. **Provenance:** `[BUILD-NEW]` `[IN-HOUSE: auth/security]` `[GAP 7.2]`.

ATM-011 — Event Bus service `[BUILD-NEW]`

- **Kind/Pri/Phase:** BUILD-NEW · P0 · 1.2. **Depends on:** ATM-002. **Blocks:** ATM-021/023/025/030.
- **Provides:** client-facing durable subscription over WS/SSE wrapping `eventbus` (NATS JetStream), sticky events + invalidation, at-least-once + durable replay. Design: build-new-subsystems.md §8.
- **Tasks:** `ATM-011.1` `Publish`; `ATM-011.2` `Subscribe(filter, sticky)` with durable JetStream consumer; `ATM-011.3` sticky last-value cache + `Invalidate`; `ATM-011.4` SSE surface (`Last-Event-ID` replay cursor); `ATM-011.5` WebSocket frame contract; `ATM-011.6` the events catalog.
- **Acceptance:** Given a disconnected client reconnecting with a cursor, When it resubscribes, Then missed events replay and sticky subjects deliver last-value first. Given an entity state change, Then the sticky value is invalidated. Given a broker restart, Then durable consumers resume with no lost event (at-least-once, idempotent consumers).

- **Test types:** unit, integration (JetStream cluster), chaos (broker restart → replay), scaling (fan-out), performance.
- **Gap:** §11. **Provenance:** `[BUILD-NEW]` `[IN-HOUSE: eventbus]` `[§3.4]`.

ATM-012 — Asset Service `[BUILD-NEW]`

- **Kind/Pri/Phase:** BUILD-NEW · P1 · 1.2. **Depends on:** ATM-003, ATM-010. **Blocks:** ATM-031/034.
- **Provides:** decoupled Catalogizer-based asset store/serve (multi-protocol, SQLCipher-at-rest, JWT+RBAC, WS, Range/`OpenSeekable`, MinIO tier, content-hash dedup, `…-web` renditions, **Proxy** links — never raw paths). Design: build-new-subsystems.md §6.
- **Tasks:** `ATM-012.1` `Put` (content-hash dedup); `ATM-012.2` `OpenRange` (Range/HLS); `ATM-012.3` `Link` (short-lived signed URL, never a path); `ATM-012.4` `Rendition` (`…-web`); `ATM-012.5` sealed AES-256-GCM dirs for sensitive assets; `ATM-012.6` asset/rendition/post_asset DDL.
- **Acceptance:** Given the same bytes twice, When `Put`, Then one row exists (dedup by content hash). Given a client, When it requests bytes, Then it receives a signed URL / Range stream, **never** a raw file path, and RBAC is enforced server-side. Given a sealed asset, Then only the service can decrypt it.
- **Test types:** unit, integration (MinIO signed URLs, Range), security (RBAC + sealed-dir), performance (streaming), scaling (50 TB+ dedup).
- **Gap:** 6.1, §11. **Provenance:** `[BUILD-NEW]` `[IN-HOUSE: Catalogizer]` `[GAP 6.1]`.

ATM-013 — `digital.vasic.auth` RS256/EdDSA + JWKS

- **Kind/Pri/Phase:** EXTEND · P1 · 1.2. **Depends on:** ATM-003. **Blocks:** ATM-010, ATM-020.
- **Provides:** asymmetric JWT signing (default is **HMAC-SHA256** `[GAP 7.2]`) + JWKS rotation.
- **Tasks:** `ATM-013.1` RS256/EdDSA signer; `ATM-013.2` JWKS endpoint + rotation; `ATM-013.3` scoped API keys; `ATM-013.4` OAuth2 external-linking.
- **Acceptance:** Given a token signed RS256, When any service verifies it, Then verification uses the published JWKS (no shared HMAC secret). Given a rotated key, Then old tokens verify until expiry and new tokens use the new key.
- **Test types:** unit, integration, security (key rotation, alg-confusion).
- **Gap:** 7.2. **Provenance:** `[EXTEND]` `[IN-HOUSE: auth]` `[GAP 7.2]`.

ATM-014 — `digital.vasic.security` searchable-yet-sealed credentials

- **Kind/Pri/Phase:** EXTEND · P2 · 1.2. **Depends on:** ATM-003.
- **Provides:** sealed key store + a **searchable-yet-sealed** credential representation (embed over a redacted/tokenized form so secrets are findable without leaking the raw value).
- **Tasks:** `ATM-014.1` sealed key store; `ATM-014.2` redacted/tokenized embedding representation; `ATM-014.3` `pkg/pii` redaction wiring.
- **Acceptance:** Given a stored credential, When a semantic search runs, Then the credential is found by meaning **without** the raw secret ever appearing in the index or logs (`[§3.6/§11.4.10]`).

- **Test types:** unit, integration, security (leak audit).
- **Gap:** 7.1. **Provenance:** `[EXTEND]` `[IN-HOUSE: security]` `[GAP 7.1]`.

ATM-015 — `Security-KMP` native secure storage

- **Kind/Pri/Phase:** HARDEN · P0 (mobile) · 1.2. **Depends on:** — . **Blocks:** ATM-046.
- **Provides:** real Android Keystore / iOS Keychain / Wasm secure storage — currently an **in-memory stub** `[GAP 7.3]` (only `DesktopSecureStorage` JVM is real AES-256-GCM).
- **Tasks:** `ATM-015.1` Android Keystore; `ATM-015.2` iOS Keychain; `ATM-015.3` Wasm/browser store; `ATM-015.4` on-device round-trip contract test.
- **Acceptance:** Given a secret stored on a real device, When the app restarts, Then it is retrieved from the platform-native secure enclave, **never** plaintext memory. **Mobile release is blocked** until this passes on-device.
- **Test types:** unit, integration (on-device), security (plaintext-at-rest negative), paired-mutation.
- **Gap:** 7.3. **Provenance:** `[HARDEN]` `[IN-HOUSE: Security-KMP]` `[GAP 7.3]` (README self-discloses stub).

3.3 Integration (sub-phase 1.3)

ATM-016 — Herald first-class Telegram thread-reader (promote MTProto)

- **Kind/Pri/Phase:** EXTEND · P0 · 1.3. **Depends on:** ATM-003. **Blocks:** ATM-017, ATM-022.
- **Provides:** promote the `qaherald` `gotd/td` MTProto user client to a first-class `channels.Channel` (forum topics via `channels.getForumTopics`, replies via `messages.getReplies`, full attachment/forward extraction), keeping the `channels.Channel` seam.
- **Tasks:** `ATM-016.1` lift the MTProto client out of `qaherald/internal/mtproto` into a first-class channel package; `ATM-016.2` implement `channels.Channel` (`SendReplyGeneric` / `BotSelfIdentity` / `DownloadAttachment` + embedded `commons.Channel`); `ATM-016.3` forum-topic + reply-thread reads; `ATM-016.4` `access_hash` resolution.
- **Acceptance:** Given a configured Telegram channel, When acquisition runs, Then full channel/group **history** is backfilled (Bot-API cannot do this); Given a forum topic, Then its topics and reply threads are read; the adapter satisfies `var _ channels.Channel`.
- **Test types:** unit, integration (live Telegram fixture — Appendix A `t.me/+622y04wzy_YzOTA0`), e2e.
- **Verified-source `[VERIFIED-SOURCE]`:** the seam is `commons_messaging/channels/channel.go`: `type Channel interface { commons.Channel; SendReplyGeneric(ctx, recipient, body, replyToID, attachments) (string, error); BotSelfIdentity(ctx) (SelfIdentity, error); DownloadAttachment(ctx, externalID, mime) (string, string, error) }`; MTProto currently lives only in `qaherald` / `docs/guides/MTPROTO.md`.
- **Gap:** 5.1. **Provenance:** `[EXTEND]` `[IN-HOUSE: herald]` `[VERIFIED-SOURCE]` `[GAP 5.1]`.

ATM-017 — ThreadReader abstraction `[BUILD-NEW]`

- **Kind/Pri/Phase:** BUILD-NEW · P1 · 1.3. **Depends on:** ATM-016. **Blocks:** ATM-018.
- **Provides:** reusable root + organic-reply assembly (exclude system replies, resolve `access_hash`), shared across messengers. Design: build-new-subsystems.md §9.
- **Tasks:** `ATM-017.1` `Assemble(ThreadRef) (Thread, error)`; `ATM-017.2` system-reply exclusion; `ATM-017.3` hashtag + attachment extraction into `Thread`.
- **Acceptance:** Given a link-only root post with tags added in a reply, When `Assemble` runs, Then the returned `Thread` contains the root + all organic replies with the reply-borne tags, and the system's own processing replies are excluded (VERIFIED §3.2.1 — critical).
- **Test types:** unit, integration (live Telegram/Max fixtures), e2e.
- **Gap:** 5.1, §11. **Provenance:** `[BUILD-NEW]` `[GAP 5.1]`.

ATM-018 — Max adapter `[BUILD-NEW]`

- **Kind/Pri/Phase:** BUILD-NEW · P0 · 1.3. **Depends on:** ATM-017. **Open:** `[OPEN: max-oneme-go-port]`.
- **Provides:** a Herald Max channel with two transports — Bot API (Go SDK, dev.max.ru) **and** a Go port of the OneMe user-WebSocket protocol for full history. Design: build-new-subsystems.md §3.
- **Tasks:** `ATM-018.1` Bot-API transport implementing `channels.Channel` (ships first); `ATM-018.2` **[spike]** OneMe user-WS protocol capture (`ATM-067`-style); `ATM-018.3` Go OneMe client (spike-gated); `ATM-018.4` ThreadReader integration.
- **Acceptance:** Given an operator Max invite link (Appendix A), When the Bot-API transport connects, Then bot-scoped posts are read and satisfy `channels.Channel`. Given the OneMe port, Then full channel/thread history reads — **but** this remains `[OPEN]` until the spike proves the protocol; it is **never** claimed working before then.
- **Test types:** unit, integration (live invite links), e2e, security (session/token via fixed Security-KMP).
- **Verified-source `[VERIFIED-SOURCE]`:** Herald has **no** `channels/max` or `channels/mtproto` package — only `docs/guides/messengers/MAX.md` (placeholder) + reserved env vars; confirms the empty stub. Reference impls `vkmax` / `max-mcp` / `MaxAPI` are Python `[RESEARCH]`.
- **Gap:** 5.1, §11. **Provenance:** `[BUILD-NEW]` `[RESEARCH]` `[VERIFIED-SOURCE]` `[GAP 5.1]`.

ATM-019 — Relational schema + migrations

- **Kind/Pri/Phase:** EXTEND · P0 · 1.3. **Depends on:** ATM-003. **Blocks:** ATM-010/020/022.
- **Provides:** the full ERD (posts, threads, replies, hashtags, categories, accounts, users, roles, permissions, memberships, assets, asset_links, processing_state, events, subscriptions, billing/metering, audit).
- **Tasks:** `ATM-019.1` post/thread/reply/hashtag/category tables; `ATM-019.2` account/user/role/permission/membership; `ATM-019.3` asset/asset_link/processing_state; `ATM-019.4` event/subscription/ billing/audit; `ATM-019.5` up/down migrations (expand-contract).
- **Acceptance:** Given the schema, When migrations apply, Then all tables + indexes exist with FK integrity; When rolled back, Then no residue. Given a post, Then its processing_state row is unique.

- **Test types:** unit, integration, migration round-trip.
- **Gap:** 3.2. **Provenance:** `[EXTEND]` `[IN-HOUSE: database]` `[§19.12]`. See ../database/index.md.

ATM-020 — REST API skeleton (`/v1` , HTTP/3 + H2)

- **Kind/Pri/Phase:** EXTEND · P0 · 1.3. **Depends on:** ATM-010, ATM-019. **Blocks:** ATM-021/044/048.
- **Provides:** `/v1` REST surface on `vasic-digital/http3` (H2 fallback) + `auth` + `ratelimiter` + `security/pkg/headers` + `middleware`.
- **Tasks:** `ATM-020.1` router + `/v1` versioning; `ATM-020.2` HTTP/3 transport + H2 fallback; `ATM-020.3` auth + rate-limit + security-headers middleware; `ATM-020.4` OpenAPI 3.1 surface + Swagger.
- **Acceptance:** Given a request, When it hits `/v1`, Then it is authn/authz-checked, rate-limited and security-headed; API p95 < 150 ms (Aggressive SLO). Given an HTTP/3-incapable client, Then it falls back to H2.
- **Test types:** unit, integration, performance (p95 < 150 ms), security (headers, rate-limit).
- **Gap:** — . **Provenance:** `[EXTEND]` `[IN-HOUSE: http3/middleware/ratelimiter]` `[Q29/Q14]`.

ATM-021 — Real-time surface (WebSocket hub + SSE)

- **Kind/Pri/Phase:** EXTEND · P1 · 1.3. **Depends on:** ATM-011, ATM-020.
- **Provides:** `digital.vasic.streaming` WS hub + SSE bridged to the Event Bus service.
- **Tasks:** `ATM-021.1` WS hub bridge; `ATM-021.2` SSE bridge; `ATM-021.3` backpressure + reconnect.
- **Acceptance:** Given a subscribed client, When an event fires, Then it is delivered over WS/SSE; Given a reconnect, Then the Event Bus replay (ATM-011) reconciles it.
- **Test types:** unit, integration, scaling (fan-out), chaos (disconnect/reconnect).
- **Gap:** 9. **Provenance:** `[EXTEND]` `[IN-HOUSE: streaming]`.

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4. Phase 2 — Processing Engine (ATM-022…ATM-043)

4.1 Content Acquisition (sub-phase 2.1)

ATM-022 — Acquisition poller + push triggers (complete-post assembly)

- **Kind/Pri/Phase:** EXTEND · P0 · 2.1. **Depends on:** ATM-016, ATM-019. **Blocks:** ATM-023 (feeds).
- **Provides:** configurable-cadence poll + push triggers; assemble the **complete post** = root + organic reply chain; store raw + emit acquisition events.
- **Tasks:** `ATM-022.1` poll scheduler (configurable cadence); `ATM-022.2` push-trigger listener; `ATM-022.3` complete-post assembly via ThreadReader; `ATM-022.4` raw persist + `post.received` emit.

- **Acceptance:** Given a new post, When either the poll or a push trigger fires, Then the complete post (root + organic replies, system replies excluded) is stored and a `post.received` event is emitted exactly once per post.
- **Test types:** unit, integration (live fixtures), e2e.
- **Gap:** 5.1. **Provenance:** `[EXTEND]` `[IN-HOUSE: herald]` `[§3.1]`.

ATM-023 — Idempotent single-claim per post

- **Kind/Pri/Phase:** EXTEND · P0 · 2.1. **Depends on:** ATM-011, ATM-019. **Blocks:** ATM-025.
- **Provides:** exactly-once processing per post on `digital.vasic.background` (Postgres row/advisory lock) even under a `post.received` event storm — the runtime analogue of §11.4.176.
- **Tasks:** `ATM-023.1` `pg_try_advisory_xact_lock`-based non-blocking claim; `ATM-023.2` processing_state transition; `ATM-023.3` storm/duplicate-delivery test harness.
- **Acceptance:** Given the same `post.received` delivered N times concurrently, When workers race, Then exactly one processes the post and the rest skip without blocking (non-blocking claim, `TryClaim` returns `false`).
- **Test types:** unit, integration, chaos (event storm), stress (N concurrent), race (`-race`).
- **Verified-source `[VERIFIED-SOURCE]`:** the exactly-once concept is now realized as a shipped module — `vasic-digital/session_orchestrator/claim/claim.go` (`Registry.TryClaim` / `Release`, `ErrNotClaimed` / `ErrClaimMismatch`) — usable directly or via the Postgres advisory-lock substrate; see ATM-062.
- **Gap:** 2.9. **Provenance:** `[EXTEND]` `[IN-HOUSE: background/session_orchestrator]` `[VERIFIED-SOURCE]`.

ATM-024 — Indirect hashtag determination

- **Kind/Pri/Phase:** BUILD-NEW · P1 · 2.1. **Depends on:** ATM-025.
- **Provides:** derive tags when absent/insufficient (torrent → `Torrent` + `ToDownload`; git host → research; YouTube/media → download+research) + AI fallback; never silently drop.
- **Tasks:** `ATM-024.1` link-class heuristics; `ATM-024.2` AI fallback classifier; `ATM-024.3` unclassified → generic-ingest + manual-review queue.
- **Acceptance:** Given a link-only post, When classified, Then the correct indirect tags are derived; Given a fully-ambiguous post, Then it is generic-ingested (stored+embedded+indexed) and queued for review — **never** dropped (`[Q32]`).
- **Test types:** unit, integration, challenges (edge inputs).
- **Gap:** — . **Provenance:** `[BUILD-NEW]` `[§3.5]`.

4.2 Processing Pipeline (sub-phase 2.2)

ATM-025 — Skill-dispatch engine `[BUILD-NEW]`

- **Kind/Pri/Phase:** BUILD-NEW · P0 · 2.2. **Depends on:** ATM-011, ATM-023, ATM-036. **Blocks:** ATM-024/026/027/028/035.
- **Provides:** hashtag/content-type → Skill(s) resolution + ordered execution (download→convert→analyze→research→reply) over BackgroundTasks with idempotent single-claim.
- **Decoupled scope:** read-only consumer of the helix_skills Skill-Graph DAG; does **not** modify the knowledge model.
- **Tasks:** `ATM-025.1` hashtag/content-type → Skill(s) resolver (reads the DAG); `ATM-025.2` step ordering by `SortOrder` + precedence; `ATM-025.3` dispatch over BackgroundTasks with single-claim; `ATM-025.4` per-step event emission.
- **Acceptance:** Given a post tagged `#Video #Research`, When dispatched, Then both the video-download Skill and the deep-research Skill run, download-first; Given a duplicate `post.received`, Then no double-dispatch; Given unknown/insufficient tags, Then fall through to ATM-024, never dropped.
- **Test types:** unit, integration, e2e, chaos, stress, performance, challenges, HelixQA.
- **Gap:** 4.1. **Provenance:** `[BUILD-NEW]` `[IN-HOUSE: helix_skills]` `[OPERATOR: all-categories-parallel]` `[GAP 4.1]`.

ATM-026 — Multi-hashtag additive resolution + precedence

- **Kind/Pri/Phase:** EXTEND · P0 · 2.2. **Depends on:** ATM-025.
- **Provides:** additive multi-Skill resolution + precedence table (download > convert > analyze > research > reply) + `SortOrder`.
- **Tasks:** `ATM-026.1` additive matcher; `ATM-026.2` precedence table; `ATM-026.3` deterministic ordering.
- **Acceptance:** Given a post with multiple categories, When resolved, Then **every** matching Skill runs, ordered deterministically so download-type Skills precede analysis-type Skills (research can consume downloaded media).
- **Test types:** unit, integration.
- **Gap:** 2.9. **Provenance:** `[EXTEND]` `[§3.3]`.

ATM-027 — Retry/back-off + circuit-breaker + soft timeout

- **Kind/Pri/Phase:** EXTEND · P0 · 2.2. **Depends on:** ATM-025.
- **Provides:** exp back-off + jitter (max 5, base 2 s, factor 2.0, cap 5 min) + circuit-breaker + per-post soft timeout (`[DEFAULT — adjustable]` 30 min research-heavy).
- **Tasks:** `ATM-027.1` retry policy (Strategy); `ATM-027.2` circuit-breaker guard; `ATM-027.3` per-post soft-timeout budget; `ATM-027.4` per-step vs whole-post retriability.

- **Acceptance:** Given a flapping external system, When it fails, Then the step retries with exp back-off up to 5 and the breaker opens on sustained failure; Given a research-heavy post exceeding its soft budget, Then it yields the slot (long downloads are delegated with callbacks).
- **Test types:** unit, integration, chaos (flapping dependency), performance.
- **Gap:** — . **Provenance:** `[EXTEND]` `[§3.3]` .

ATM-028 — Download Manager `[BUILD-NEW]`

- **Kind/Pri/Phase:** BUILD-NEW · P0 · 2.2. **Depends on:** ATM-029, ATM-030. **Blocks:** ATM-012.
- **Provides:** multi-protocol download engine (HTTP/1.1/2/3+QUIC+Brotli via `http3` ; FTP/SMB/NFS/WebDav by reusing `filesystem`) with queue, resumable+segmented, progress, retry, standardized callback. Design: build-new-subsystems.md §5.
- **Tasks:** `ATM-028.1` HTTP/3 source adapter; `ATM-028.2` filesystem source adapter (FTP/SMB/NFS/WebDav); `ATM-028.3` segmented-transfer engine + resume from persisted offset; `ATM-028.4` bounded worker-pool queue; `ATM-028.5` callback emitter (shared schema, ATM-030).
- **Acceptance:** Given a large HTTP/3 URL, When enqueued, Then it downloads in N segments with progress; Given a mid-transfer kill, Then it resumes from the last committed offset; Given completion, Then a `JobUpdate{succeeded, asset_ref}` callback fires idempotently.
- **Test types:** unit, integration (each protocol vs a real server), stress (100 concurrent), chaos (mid-transfer kill → resume), performance (throughput).
- **Gap:** 6.2, 6.3, §11. **Provenance:** `[BUILD-NEW]` `[IN-HOUSE: filesystem/http3]` `[GAP 6.3]` .

ATM-029 — `filesystem` HTTP source + NFS platform-listing fix

- **Kind/Pri/Phase:** EXTEND · P1 · 2.2. **Depends on:** — . **Blocks:** ATM-028.
- **Provides:** add an HTTP(S) source protocol to `digital.vasic.filesystem` ; fix NFS non-Linux listing; expose `OpenSeekable` uniformly for Range.
- **Tasks:** `ATM-029.1` HTTP(S) source; `ATM-029.2` NFS `SupportedProtocols` platform gate; `ATM-029.3` uniform `OpenSeekable` .
- **Acceptance:** Given an HTTP(S) URL, When opened, Then `OpenSeekable` returns a Range-capable reader; Given a non-Linux host, Then NFS is **not** advertised in `SupportedProtocols` (it errors today yet is still listed).
- **Test types:** unit, integration (HTTP/FTP/SMB/NFS/WebDav), cross-platform.
- **Gap:** 6.2. **Provenance:** `[EXTEND]` `[IN-HOUSE: filesystem]` `[GAP 6.2]` .

ATM-030 — Standardized callback/task module `[BUILD-NEW]`

- **Kind/Pri/Phase:** BUILD-NEW · P1 · 2.2. **Depends on:** ATM-011. **Blocks:** ATM-028/031/032.
- **Provides:** one canonical async-job contract (job id, state, progress, result/asset ref, error) applied to Boba, MeTube, Download Manager. Design: build-new-subsystems.md §2.

- **Tasks:** `ATM-030.1` `JobUpdate` / `JobState` / `JobError` schema; `ATM-030.2` `TaskProducer` + `UpdateSink`; `ATM-030.3` HMAC-signed idempotent webhook POST + exp back-off; `ATM-030.4` Event Bus publish sink.
- **Acceptance:** Given a completed task, When delivery runs, Then an idempotent HMAC-signed POST (or Event Bus publish) carries the `JobUpdate`; Given a dropped webhook, Then it redelivers under back-off and consumers treat duplicates idempotently (at-least-once).
- **Test types:** unit, integration (real Boba/MeTube), chaos (webhook drop/redeliver), stress, security (HMAC + SSRF).
- **Gap:** 6.6, §11. **Provenance:** `[BUILD-NEW]` `[GAP 6.6]`.

ATM-031 — MeTube outbound completion webhook

- **Kind/Pri/Phase:** EXTEND · P0 · 2.2. **Depends on:** ATM-012, ATM-030.
- **Provides:** add an **outbound completion webhook** to MeTube (poll-only today `[GAP 6.5]`) + Asset Service integration + `…-web` conversion profiles.
- **Tasks:** `ATM-031.1` outbound webhook on the postprocessor sidecar; `ATM-031.2` Asset Service handoff; `ATM-031.3` `…-web` conversion-profile config.
- **Acceptance:** Given a completed MeTube download, When post-processing finishes, Then it **pushes** a shared-schema completion webhook (no polling) and the asset lands in the Asset Service with a `…-web` rendition.
- **Test types:** unit, integration (real MeTube), chaos (webhook retry).
- **Gap:** 6.5. **Provenance:** `[EXTEND]` `[IN-HOUSE: MeTube]` `[GAP 6.5]`.

ATM-032 — Boba callback standardization

- **Kind/Pri/Phase:** EXTEND · P1 · 2.2. **Depends on:** ATM-030.
- **Provides:** standardize Boba's existing SSE + `POST /api/v1/hooks` to the shared callback schema; Asset Service integration.
- **Tasks:** `ATM-032.1` map SSE `result_found` → `JobUpdate`; `ATM-032.2` map `POST /api/v1/hooks` to the shared schema; `ATM-032.3` Asset Service handoff.
- **Acceptance:** Given a Boba torrent result, When found, Then the callback conforms to the shared schema (job id/state/progress/asset ref/error) and the asset lands in the Asset Service.
- **Test types:** unit, integration (real Boba).
- **Gap:** 6.4. **Provenance:** `[EXTEND]` `[IN-HOUSE: Boba]` `[GAP 6.4]`.

ATM-033 — OCR adapter `[BUILD-NEW]`

- **Kind/Pri/Phase:** BUILD-NEW · P0 · 2.2. **Depends on:** — (independent).
- **Provides:** a first-class `OCRProvider` seam behind VisionEngine (Tesseract via cgo `gosseract` / subprocess + PaddleOCR option), per-word boxes, offline+deterministic, then a hybrid Tesseract→ LLM-vision pipeline. Design: build-new-subsystems.md §4.

- **Tasks:** `ATM-033.1` `OCRProvider` interface + factory; `ATM-033.2` Tesseract adapter (cgo `-tags ocr` + subprocess fallback); `ATM-033.3` PaddleOCR adapter; `ATM-033.4` hybrid escalation of low-confidence words to LLM-vision; `ATM-033.5` wire `TextRegion` to the real recognizer.
- **Acceptance:** Given a `#Comic` page, When OCR runs, Then a full transcription with per-word boxes is produced offline+deterministically and semantically ingested (no research/Skills); Given an ambiguous region, Then only that region escalates to LLM-vision.
- **Test types:** unit, integration (real images), benchmarking (throughput), UX (accuracy), challenges.
- **Gap:** 2.6, §11. **Provenance:** `[BUILD-NEW]` `[IN-HOUSE: VisionEngine]` `[GAP 2.6]`.

ATM-034 — Media transcode profiles

- **Kind/Pri/Phase:** EXTEND · P1 · 2.2. **Depends on:** ATM-012, ATM-031.
- **Provides:** raw preserved + `…-web` H.264/AAC fMP4 (+H.265/AV1); 1080/720/480 HLS+DASH; audio MP3 320k/Opus 128k/FLAC.
- **Tasks:** `ATM-034.1` `…-web` H.264/AAC fMP4; `ATM-034.2` H.265/AV1 renditions; `ATM-034.3` adaptive HLS+DASH ladders; `ATM-034.4` audio profiles.
- **Acceptance:** Given a raw video asset, When transcoded, Then the raw is preserved and a `…-web` rendition + adaptive HLS/DASH ladder exist; Given a broken physical link, Then it is re-downloadable via REST (re-invoke Download Manager).
- **Test types:** unit, integration, performance (transcode throughput), UX (playback).
- **Gap:** 6.1. **Provenance:** `[EXTEND]` `[Q36]`.

ATM-035 — Post-processing (status reply, events, embed+index, Skill-Tree growth)

- **Kind/Pri/Phase:** EXTEND · P0 · 2.2. **Depends on:** ATM-025, ATM-039.
- **Provides:** status reply to the original post (Robot/User acct), fire events, embed+index generated materials, grow Skill-Trees; **never process own replies**.
- **Tasks:** `ATM-035.1` status-reply composer + poster; `ATM-035.2` result events; `ATM-035.3` embed
    - semantic index of generated materials; `ATM-035.4` Skill-Tree growth; `ATM-035.5` self-reply guard.
- **Acceptance:** Given a processed post, When post-processing runs, Then a status reply (metrics + asset refs) is posted from the Robot/User account, events fire, generated materials are embedded + indexed, and Skill-Trees grow; Given the system's own reply, Then it is **never** processed.
- **Test types:** unit, integration, e2e, challenges.
- **Gap:** — . **Provenance:** `[EXTEND]` `[§3.2.3]`.

4.3 Skills Integration (sub-phase 2.3)

ATM-036 — Author a Skill (recipe) for every documented content type

- **Kind/Pri/Phase:** EXTEND · P0 · 2.3. **Depends on:** — . **Blocks:** ATM-025 (needs recipes).

- **Provides:** a Skill per content type — all 18 categories in parallel `[OPERATOR]` (Video, Torrent, Series, Movie, Research, Documentary, Concert, Game, Software, Channel, Playlist, Music, Book, Comic, Netflix, Training, Technology, Screenshot/QR-sensitive).
- **Tasks:** one task per category cluster, e.g. `ATM-036.1` download-family (Video/Torrent/Series/ Movie/ Netflix/Music/Concert); `ATM-036.2` research-family (Research/Technology/Documentary/Training/ Book-IT); `ATM-036.3` media-analysis (Comic/Screenshot/QR); `ATM-036.4` collection (Channel/ Playlist/ Game/Software).
- **Acceptance:** Given each documented content type, When a matching post arrives, Then its Skill recipe resolves and orders the correct steps (e.g. Playlist groups with ordering-number prefixes for watch order; Comic → OCR transcription + semantic ingest, no research).
- **Test types:** unit, integration, e2e, challenges (per-category scenario banks).
- **Gap:** 4.1. **Provenance:** `[EXTEND]` `[IN-HOUSE: helix_skills]` `[OPERATOR]` `[§3.2.2]`.

ATM-037 — Standardize the Skill file format

- **Kind/Pri/Phase:** HARDEN · P1 · 2.3. **Depends on:** ATM-036.
- **Provides:** one canonical `SKILL.md` YAML-frontmatter schema; migrate no-frontmatter dirs; fix the `catalog-engine` “Test Catalog” mislabel; deterministic `INDEX.md` regen.
- **Tasks:** `ATM-037.1` canonical `SKILL.md` schema (`name` / `description` / `version`); `ATM-037.2` migrate lowercase no-frontmatter `skill.md` dirs; `ATM-037.3` fix the catalog mislabel; `ATM-037.4` deterministic `INDEX.md` regeneration.
- **Acceptance:** Given the skills tree, When validated, Then every skill uses the canonical `SKILL.md` frontmatter and `INDEX.md` regenerates identically across runs.
- **Test types:** unit, integration, challenges (format lint).
- **Verified-source `[VERIFIED-SOURCE]`:** the inconsistency is real — `constitution/skills/workable-item-lifecycle/SKILL.md` carries YAML frontmatter (`name` / `description` / `version: 1.0.0`) while `constitution/skills/media-validator/skill.md` has **no** frontmatter (starts `# Media Validator Skill`). `docs/catalog/catalog.json` is titled “Helix Skills — **Test Catalog**” and its entries are grammar/mutation *tests*, not skills (the mislabel).
- **Gap:** 4.1. **Provenance:** `[HARDEN]` `[IN-HOUSE: helix_skills]` `[VERIFIED-SOURCE]` `[GAP 4.1]`.

ATM-038 — Triage the Skill-Graph MVP open findings

- **Kind/Pri/Phase:** AUDIT · P1 · 2.3. **Depends on:** ATM-037.
- **Provides:** burn down the Skill-Graph MVP’s 95 open findings (of 136) relevant to Thready.
- **Tasks:** `ATM-038.1` triage the 95 open findings for Thready relevance; `ATM-038.2` fix the Thready-relevant subset; `ATM-038.3` reconcile the unreliable generated `INDEX.md` /README counts.
- **Acceptance:** Given the findings register, When triaged, Then every Thready-relevant finding is fixed or explicitly deferred with rationale; counts regenerate deterministically.
- **Test types:** unit, integration, AUDIT.

- **Gap:** 4.1. **Provenance:** `[AUDIT]` `[IN-HOUSE: helix_skills]` `[GAP 4.1]`.

4.4 Semantic Search (sub-phase 2.4)

ATM-039 — Semantic-search service `[BUILD-NEW]`

- **Kind/Pri/Phase:** BUILD-NEW · P0 · 2.4. **Depends on:** ATM-004, ATM-040. **Blocks:** ATM-035.
- **Provides:** Lumen-style in-house search — chunk (AST/tree-sitter for code, Markdown/structured for docs) → `embeddings` → `vectordb` → `MCP_Module`, over HelixLLM `/v1/embeddings`; indexes posts + generated materials; hydrates ids from the relational store. Design: build-new-subsystems.md §10.
- **Tasks:** `ATM-039.1` chunkers (AST + Markdown); `ATM-039.2` `Index` (embed+upsert); `ATM-039.3` `Query` (cosine top-k, < 500 ms); `ATM-039.4` `MCP_Module` tool exposure; `ATM-039.5` the anti-bluff paired-mutation gate proving real relevance.
- **Acceptance:** Given indexed posts + generated materials, When a query runs, Then cosine top-k hits hydrate against relational rows in < 500 ms; Given `HELIX_EMBEDDING_PROVIDER != llama`, Then the service **refuses** (503) rather than returning hash noise.
- **Test types:** unit, integration (real embeddings + pgvector), performance (< 500 ms), benchmarking, challenges (relevance), paired-mutation.
- **Gap:** 2.7, §11. **Provenance:** `[BUILD-NEW]` `[IN-HOUSE: embeddings/vectordb/rag]` `[GAP 2.1/2.7]`.

ATM-040 — HelixLLM: enforce `llama` embedder, fail loudly on `HashEmbedder`

- **Kind/Pri/Phase:** HARDEN · P0 · 2.4. **Depends on:** — . **Blocks:** ATM-039, ATM-004.4, ATM-041.
- **Provides:** enforce `HELIX_EMBEDDING_PROVIDER=llama`; **fail loudly** (not warn) if the non-semantic `HashEmbedder` is selected in a RAG/search context; load a real code-tuned embedding GGUF.
- **Tasks:** `ATM-040.1` provider guard rejecting `hash`/`local`/unrecognised in semantic contexts; `ATM-040.2` load a real embedding GGUF; `ATM-040.3` startup fail-loud (not the current WARN).
- **Acceptance:** Given `NewEmbedder("hash"|"local"|<unknown>, …)` in a semantic context, When invoked, Then it errors and refuses to serve; Given `NewEmbedder("llama", …)`, Then it serves real vectors. A paired-mutation gate proves the guard actually fires.
- **Test types:** unit, integration, paired-mutation (anti-bluff), security.
- **Verified-source** `[VERIFIED-SOURCE]`: `HelixLLM/internal/knowledge/embedding_providers.go`: `func NewEmbedder(provider, apiKey, model string, dimension int) (Embedder, error)` — `"llama"` → `LlamaEmbedder`; `"hash"/"local"` **and any unrecognised name** → `NewHashEmbedder(dimension)` (the exact silent-fallback trap). Companion `cmd/helixllm/embedder_warning_test.go` shows today's behaviour is a warning, not a hard fail.
- **Gap:** 2.1. **Provenance:** `[HARDEN]` `[IN-HOUSE: HelixLLM]` `[VERIFIED-SOURCE]` `[GAP 2.1]`.

ATM-041 — `embeddings`: native llama.cpp/HelixLLM provider + dimension discovery

- **Kind/Pri/Phase:** EXTEND · P1 · 2.4. **Depends on:** ATM-040.
- **Provides:** a first-class llama.cpp/HelixLLM embeddings provider (not a repurposed OpenAI client) with health checks + dimension discovery; parameterize embedding dimension (remove the 768 hardcode).
- **Tasks:** `ATM-041.1` native provider; `ATM-041.2` `/v1/embeddings` OpenAI-shape contract test (sort by `index`); `ATM-041.3` config-driven dimension validated against the model.
- **Acceptance:** Given a real llama embedding model, When the provider queries it, Then dimensions are discovered and validated (no 768 hardcode) and `/v1/embeddings` returns `{data: [{embedding,index}]}`.
- **Test types:** unit, integration (real embeddings), contract.
- **Gap:** 2.1, 2.7. **Provenance:** `[EXTEND]` `[IN-HOUSE: embeddings]` `[GAP 2.7]`.

ATM-042 — `vectordb`: harden the Qdrant backend to pgvector parity

- **Kind/Pri/Phase:** HARDEN · P1 · 2.4. **Depends on:** ATM-004.
- **Provides:** Qdrant backend to full pgvector parity behind `VectorStore` + ANN index tuning + benchmark for < 500 ms SLO.
- **Tasks:** `ATM-042.1` Qdrant end-to-end integration; `ATM-042.2` ANN index tuning; `ATM-042.3` < 500 ms benchmark; `ATM-042.4` config-swap parity test vs pgvector.
- **Acceptance:** Given the same corpus, When queried on Qdrant vs pgvector, Then results are equivalent and Qdrant meets the < 500 ms SLO; Thready can swap backend by config.
- **Test types:** unit, integration (real Qdrant), performance, benchmarking.
- **Gap:** 3.1. **Provenance:** `[HARDEN]` `[IN-HOUSE: vectordb]` `[GAP 3.1]`.

ATM-043 — `database`: time-partitioning + retention/archive helpers

- **Kind/Pri/Phase:** EXTEND · P1 · 2.4. **Depends on:** ATM-003.
- **Provides:** time-partitioning + retention/archive helpers for 10k+/day posts; validate `storage` MinIO signed-URL parity; pool + pgvector co-location tuning.
- **Tasks:** `ATM-043.1` time-partitioned posts; `ATM-043.2` retention/archive helpers (per-account overrides); `ATM-043.3` MinIO signed-URL parity vs CloudFront; `ATM-043.4` pool + co-location tuning.
- **Acceptance:** Given 10k+/day posts, When partitioning + retention run, Then old partitions archive per the per-account retention policy and queries stay within SLO; MinIO signed URLs match the CloudFront contract.
- **Test types:** unit, integration (MinIO), performance, scaling.
- **Gap:** 3.2. **Provenance:** `[EXTEND]` `[IN-HOUSE: database/storage]` `[GAP 3.2]`.

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5. Phase 3 — Client Applications (ATM-044…ATM-051)

ATM-044 — Angular 19 product portal

- **Kind/Pri/Phase:** EXTEND · P0 · 3.1. **Depends on:** ATM-020, ATM-060. **Provides:** the Web portal on the shared OpenDesign `design_system` (Web + CLI first `[OPERATOR]`), i18n via HelixTranslate.
- **Tasks:** `ATM-044.1` scaffold on `design_system` + Material 19; `ATM-044.2` auth/session flows; `ATM-044.3` post/thread/asset views; `ATM-044.4` search UI (`/v1/search`); `ATM-044.5` ngx-translate.
- **Acceptance:** Given an authenticated user, When they open the portal, Then posts/threads/assets/ search render on the design system; page load < 1.5 s (Aggressive SLO).
- **Test types:** unit (Jasmine+Karma), e2e (Cypress + cypress-axe), performance, UX.
- **Gap:** 8.1. **Provenance:** `[EXTEND]` `[IN-HOUSE: design_system]` `[Q13/Q17]`.

ATM-045 — Tauri 2 desktop

- **Kind/Pri/Phase:** EXTEND · P1 · 3.2. **Depends on:** ATM-044. **Provides:** Tauri 2 desktop (Rust core + Angular UI) — org standard.
- **Tasks:** `ATM-045.1` Tauri shell wrapping the Angular UI; `ATM-045.2` native integration; `ATM-045.3` packaging.
- **Acceptance:** Given the desktop build, When launched, Then it wraps the portal natively and shares the Angular UI; `cargo clippy -D warnings` clean.
- **Test types:** unit, integration, e2e.
- **Gap:** — . **Provenance:** `[EXTEND]` `[IN-HOUSE]` `[Q18]`.

ATM-046 — Native mobile clients + KMP shared logic

- **Kind/Pri/Phase:** BUILD-NEW · P2 · 3.3. **Depends on:** ATM-047, ATM-015. **Provides:** native clients (Compose/Android, SwiftUI/iOS, ArkTS/HarmonyOS, Qt/Aurora + `helix_shims`) + KMP shared logic.
- **Tasks:** `ATM-046.1` Compose/Android; `ATM-046.2` SwiftUI/iOS; `ATM-046.3` ArkTS/HarmonyOS; `ATM-046.4` Qt/Aurora + shims; `ATM-046.5` KMP shared-logic wiring.
- **Acceptance:** Given the fixed Security-KMP (ATM-015), When a mobile client stores tokens, Then they use native Keychain/KeyStore; the ArkTS + Qt clients reach HarmonyOS + Aurora (native is the only path). **Mobile release is blocked until ATM-015 passes on-device.**
- **Test types:** unit, integration, e2e, security (on-device storage).
- **Gap:** 8.3, 8.4, 8.5. **Provenance:** `[BUILD-NEW]` `[IN-HOUSE: *-KMP/helix_shims]` `[Q19]` `[GAP 8.5]`.

ATM-047 — KMP module fleet: CI + Maven publish + Database-KMP SQLDelight

- **Kind/Pri/Phase:** HARDEN · P1 · 3.x. **Depends on:** — . **Blocks:** ATM-046.

- **Provides:** CI + Maven publish + shared convention plugin for the 10 `*-KMP` modules; implement `Database-KMP` with SQLDelight (interfaces-only today `[GAP 8.4]`).
- **Tasks:** `ATM-047.1` shared convention plugin; `ATM-047.2` CI + Maven publish; `ATM-047.3` `Database-KMP` SQLDelight implementation.
- **Acceptance:** Given a KMP module, When CI runs, Then it builds/tests and publishes to Maven; `Database-KMP` has a working SQLDelight implementation (not just interfaces).
- **Test types:** unit (Kotest), integration, instrumented (Compose).
- **Gap:** 8.4. **Provenance:** `[HARDEN]` `[IN-HOUSE: *-KMP]` `[GAP 8.4]`.

ATM-048 — Go/Cobra CLI (headless, shares SDK with TUI)

- **Kind/Pri/Phase:** EXTEND · P0 · 3.4. **Depends on:** ATM-060. **Provides:** the headless Cobra CLI sharing the SDK — first alongside Web `[OPERATOR]`.
- **Tasks:** `ATM-048.1` Cobra command tree; `ATM-048.2` SDK wiring; `ATM-048.3` MCP tool for search.
- **Acceptance:** Given the CLI, When a command runs, Then it uses the generated SDK and can trigger `/v1/search` (+ MCP tool for CLI agents).
- **Test types:** unit, integration, e2e.
- **Gap:** — . **Provenance:** `[EXTEND]` `[IN-HOUSE]` `[Q13/§21.3]`.

ATM-049 — Bubble Tea + Cobra + Lipgloss TUI

- **Kind/Pri/Phase:** EXTEND · P1 · 3.4. **Depends on:** ATM-048. **Provides:** the TUI (as `helix_track_cli`).
- **Tasks:** `ATM-049.1` Bubble Tea model; `ATM-049.2` Lipgloss styling on the design tokens; `ATM-049.3` shared SDK/commands with the CLI.
- **Acceptance:** Given the TUI, When launched, Then it renders the same SDK-backed operations as the CLI with a Bubble Tea UI.
- **Test types:** unit, integration, UX.
- **Gap:** 8.6. **Provenance:** `[EXTEND]` `[IN-HOUSE: helix_track_cli]` `[Q20]`.

ATM-050 — Thready brand theme + mature design_system + publish npm

- **Kind/Pri/Phase:** EXTEND · P1 · 3.5. **Depends on:** — . **Provides:** Thready brand theme from `Logo.png` on the helix-green base; mature `design_system`; publish `@vasic-digital/design-system`.
- **Tasks:** `ATM-050.1` brand theme tokens; `ATM-050.2` light+dark variants; `ATM-050.3` npm publish.
- **Acceptance:** Given the design system, When themed, Then Thready branding (helix-green base) renders in light+dark and the package publishes to npm.
- **Test types:** unit, integration, visual-regression.
- **Gap:** 8.1. **Provenance:** `[EXTEND]` `[IN-HOUSE: design_system]` `[GAP 8.1]`.

ATM-051 — helix_design per-platform token packages

- **Kind/Pri/Phase:** HARDEN · P1 · 3.5. **Depends on:** ATM-050. **Provides:** per-platform token packages (Flutter, Qt/Aurora, CSS) from the OpenDesign source (currently a scaffold `[GAP 8.2]`).
- **Tasks:** `ATM-051.1` Flutter tokens; `ATM-051.2` Qt/Aurora tokens; `ATM-051.3` CSS tokens.
- **Acceptance:** Given the OpenDesign source, When token packages build, Then Flutter/Qt/CSS consumers render consistent tokens in lockstep with `design_system`.
- **Test types:** unit, integration, visual-regression.
- **Gap:** 8.2. **Provenance:** `[HARDEN]` `[IN-HOUSE: helix_design]` `[GAP 8.2]`.

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6. Phase 4 — Testing & Deployment (ATM-052…ATM-057)

ATM-052 — Unit + mutation testing + paired-mutation anti-bluff gates

- **Kind/Pri/Phase:** EXTEND · P0 · 4.1. **Depends on:** per-item. **Provides:** unit + `go-mutesting`
  - paired-mutation gates proving real behavior, not stubs.
- **Tasks:** `ATM-052.1` per-package unit + `-race`; `ATM-052.2` `go-mutesting` runs; `ATM-052.3` paired-mutation gates for every scaffold dependency.
- **Acceptance:** Given a scaffold dependency, When the paired-mutation gate runs, Then mutating the impl makes the test FAIL and restoring makes it PASS (the test detects real behavior).
- **Test types:** unit, mutation, paired-mutation.
- **Gap:** 12. **Provenance:** `[EXTEND]` `[CONST-035]` `[GAP 12]`.

ATM-053 — Integration (cross-module contracts + critical paths)

- **Kind/Pri/Phase:** EXTEND · P0 · 4.2. **Depends on:** Phase 1–2. **Provides:** all cross-module contracts + critical paths first; no-fakes-beyond-unit.
- **Tasks:** `ATM-053.1` critical-path contracts; `ATM-053.2` expand to full combinations; `ATM-053.3` real-dependency harness (no mocks beyond unit).
- **Acceptance:** Given two modules with a contract, When integration runs, Then they interoperate against the real system (mocks unit-only, `[§11.4.27]`).
- **Test types:** integration, contract.
- **Gap:** §11.4.27. **Provenance:** `[EXTEND]` `[Q24]`.

ATM-054 — System/e2e + HelixQA banks + Challenges banks

- **Kind/Pri/Phase:** EXTEND · P0 · 4.3. **Depends on:** ATM-053. **Provides:** system/e2e + HelixQA YAML banks (runtime evidence) + Challenges scenario banks + VR family + DocProcessor coverage.

- **Tasks:** `ATM-054.1` HelixQA YAML banks per service+client; `ATM-054.2` Challenges scenario banks; `ATM-054.3` visual-regression (Panoptic/VR) scenarios; `ATM-054.4` DocProcessor feature-map coverage.
- **Acceptance:** Given each feature × platform, When the banks run, Then runtime evidence proves the behavior (anti-bluff) and docs↔tests coverage is tracked.
- **Test types:** e2e, system, HelixQA, challenges, visual-regression.
- **Gap:** 9.1, 9.3, 9.4. **Provenance:** `[EXTEND]` `[IN-HOUSE: HelixQA/challenges/Panoptic/DocProcessor]`.

ATM-055 — Security (SonarQube + Snyk + fuzzing + CVE + DDoS/authz)

- **Kind/Pri/Phase:** EXTEND · P0 · 4.4. **Depends on:** ATM-053. **Provides:** SonarQube (CLI + rootless-Podman server) + Snyk + fuzzing + CVE + DDoS/authz.
- **Tasks:** `ATM-055.1` SonarQube CLI + server; `ATM-055.2` Snyk; `ATM-055.3` fuzzing; `ATM-055.4` CVE scanning; `ATM-055.5` DDoS/authz/pen tests (via HelixQA security).
- **Acceptance:** Given the codebase, When security scans run, Then SonarQube/Snyk gates pass, fuzzers find no crashers, and the authz matrix resists privilege escalation.
- **Test types:** security, fuzzing, penetration.
- **Gap:** 7.1. **Provenance:** `[EXTEND]` `[CONSTITUTION §11.4.184]` `[Q23]`.

ATM-056 — Performance/benchmarking/stress/scaling/chaos + RPO/RTO

- **Kind/Pri/Phase:** EXTEND · P1 · 4.5. **Depends on:** ATM-053. **Provides:** validate SLOs (API p95 < 150 ms, search < 500 ms) + RPO/RTO.
- **Tasks:** `ATM-056.1` API p95 benchmark; `ATM-056.2` search < 500 ms benchmark; `ATM-056.3` stress/scaling; `ATM-056.4` chaos + RPO ≈ 1 h / RTO ≈ 4 h validation.
- **Acceptance:** Given the deployed system, When load/chaos tests run, Then the Aggressive SLOs hold and a simulated disaster restores within RPO ≈ 1 h / RTO ≈ 4 h.
- **Test types:** performance, benchmarking, stress, scaling, chaos.
- **Gap:** 3.1, 3.2. **Provenance:** `[EXTEND]` `[Q14/Q45]`.

ATM-057 — Production deployment (3 isolated env stacks)

- **Kind/Pri/Phase:** EXTEND · P0 · 4.6. **Depends on:** ATM-002, ATM-006. **Provides:** 3 isolated env stacks (`dev.` / `sta.` / `thready.`), rootless Podman Compose, Let's Encrypt, backup/DR runbook.
- **Tasks:** `ATM-057.1` three isolated compose stacks behind subdomains; `ATM-057.2` per-subdomain certs; `ATM-057.3` backup (daily full + hourly DB incrementals); `ATM-057.4` restore runbook (PITR).
- **Acceptance:** Given the Hetzner host, When deployed, Then `dev.` / `sta.` / `thready.` run as fully isolated container stacks with their own certs, and the documented restore runbook meets RPO/RTO.
- **Test types:** integration, e2e, chaos (DR drill), performance.

- **Gap:** — . **Provenance:** `[EXTEND]` `[OPERATOR]` `[Q8/Q41/Q45]`. See ../deployment/index.md.

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7. Cross-cutting (ATM-058…ATM-072)

These span all phases. Full cards for the items whose scope changed in Pass 3; the rest carry the summary from workable-items.md §7.

ATM-058 — Anti-bluff sweep

- **Kind/Pri:** AUDIT · P1. **Provides:** a paired-mutation gate for every `SCAFFOLD` / `DESIGN-ONLY` dep before reliance. **Acceptance:** Given any scaffold dep (HelixLLM embedder, Security-KMP, TOON/ token_optimizer), When the gate runs, Then a real-behavior mutation is proven detected before the dep is relied on. **Test types:** paired-mutation, integration. **Gap:** 12.

ATM-059 — Decoupling audit `[§11.4.28]`

- **Kind/Pri:** AUDIT · P1. **Provides:** each reused submodule verified project-not-aware/config- injected; no nested own-org chains. **Acceptance:** Given each reused module, When audited, Then it is config-injected with no hardcoded Thready specifics and no forbidden nested own-org submodule chain. **Test types:** AUDIT, integration. **Gap:** 12.

ATM-060 — SDK codegen (OpenAPI 3.1 + Protobuf via helix_proto)

- **Kind/Pri:** BUILD-NEW · P1. **Depends on:** ATM-020 (REST surface). **Blocks:** ATM-044/048.
- **Provides:** OpenAPI 3.1 + Protobuf via the `helix_proto` pattern (`buf` → Go/Rust; `openapi-generator` → TS/Dart) + a thin idiomatic layer per language. **Tasks:** `ATM-060.1` OpenAPI 3.1 surface; `ATM-060.2` Protobuf DTO/event contracts; `ATM-060.3` `buf` Go/Rust codegen; `ATM-060.4` `openapi-generator` TS/Dart; `ATM-060.5` thin idiomatic layer + registry publish.
- **Acceptance:** Given the OpenAPI+Protobuf contracts, When codegen runs, Then versioned SDKs generate for Go/Rust/TS/Dart with a thin hand-written idiomatic layer, tested + registry-published.
- **Test types:** unit, integration, contract. **Gap:** 13. **Provenance:** `[BUILD-NEW]` `[IN-HOUSE: helix_proto]` `[§13.1]`.

ATM-061 — TOON + token_optimizer: implement or scope-out

- **Kind/Pri:** HARDEN · P1. **Provides:** implement `TOON` (SCAFFOLD, `ErrTOONEncodingNotImplemented`)
    - `token_optimizer` (partial, only `pkg/config`) **or** explicitly mark out-of-scope for MVP. Until then token optimization relies on the **toolkit's** working `toon.mjs` / `toon_encode.py`, not these modules. **Acceptance:** Given the token-optimization mandate `[§11.4.198]`, When the decision is recorded, Then either both modules are implemented+tested or MVP explicitly uses the toolkit

utilities and marks the modules out-of-scope — **never** presenting TOON-the-module as working. **Test types:** unit, integration, paired-mutation. **Gap:** 2.9.

ATM-062 — session_orchestrator: wire the (now-implemented) claim registry

- **Kind/Pri:** ~~BUILD-NEW~~ **HARDEN/WIRE** (rescoped Pass 3) · P1. **Provides:** the atomic track-claim registry §11.4.176(A). **Pass-3 update:** the gap register flagged this **DESIGN-ONLY**, but at source `vasic-digital/session_orchestrator` now ships **real Go code** — so the item is **rescoped from build-from-scratch to wire+verify+integrate**.
- **Tasks:** `ATM-062.1` verify the `claim` package against §11.4.176(A) semantics; `ATM-062.2` wire it as the dev-fleet claim registry (replacing the interim advisory-lock realization); `ATM-062.3` reuse the same registry for the runtime single-claim (ATM-023); `ATM-062.4` contract + paired-mutation tests.
- **Acceptance:** Given two tracks claiming the same key, When both call `TryClaim`, Then exactly one wins and the other gets the unchanged live claim (exactly-once); `Release` with a mismatched id returns `ErrClaimMismatch`.
- **Verified-source `[VERIFIED-SOURCE]`:** `vasic-digital/session_orchestrator/claim/claim.go` — `type Registry struct { claims map[string]Claim … }`, `func New(cfg Config) *Registry`, `TryClaim(...)` returning an `Outcome` (never-erroring), `Release` returning `ErrNotClaimed` / `ErrClaimMismatch`; sibling packages `alias/`, `scheduler/`, `supervisor/`, `claim/select.go`. Package doc: "the exactly-once claim + deadlock-free device-lock" (WS-D DESIGN §1.3–§1.4). This **supersedes** the DESIGN-ONLY claim in the gap register `[GAP 2.9]`.
- **Test types:** unit, integration, paired-mutation, race. **Gap:** 2.9. **Provenance:** `[HARDEN]` `[IN-HOUSE: session_orchestrator]` `[VERIFIED-SOURCE]`.

ATM-063 — HelixAgent identity/module split + close admitted stubs

- **Kind/Pri:** HARDEN · P2. **Provides:** resolve the HelixAgent↔HelixCode identity/module split; enumerate + close admitted stub/bluff areas; pin only the packages Thready needs (ensemble/debate). **Acceptance:** Given HelixAgent, When Thready pins it, Then only the needed ensemble/debate packages are depended on and their real coverage is verified (not the 65.6% claim taken on faith). **Test types:** AUDIT, integration. **Gap:** 2.2.

ATM-064 — LLMProvider per-adapter contract tests

- **Kind/Pri:** AUDIT · P1. **Provides:** per-adapter contract tests for every adapter Thready relies on; confirm whether the embeddings interface lives here or in `Embeddings`. **Acceptance:** Given each relied-on adapter, When contract tests run, Then retry/circuit-breaker/health behave as specified and the embeddings-interface location is confirmed. **Test types:** contract, integration. **Gap:** 2.3, 13.

ATM-065 — LLMsVerifier port reconciliation + stable scoring API

- **Kind/Pri:** HARDEN · P2. **Provides:** reconcile the `:7061` vs `:8080` port; expose a stable scoring API for HelixLLM's fallback chain; prune the `FINAL_*` / `COMPLETE_*` / `ULTIMATE_*` doc sprawl. **Acceptance:**

Given HelixLLM's fallback chain, When it scores models, Then a single-source-of-truth port + stable scoring API are used. **Test types:** integration, contract. **Gap:** 2.5.

ATM-066 — Constitution anchor verification `[OPEN]`

- **Kind/Pri:** AUDIT · P1. **`[OPEN: constitution-anchor-verify]`** — the local Constitution copy tops out at §11.4.192; the exact normative text of **§11.4.196/198/209** was taken from the final request's descriptions, not read at source. **Resolution:** re-verify against the canonical `HelixDevelopment/helix_constitution` and update citations. **Pass-3 check:** the constitution repo exists (`HelixDevelopment/helix_constitution`, public) but the specific §-anchors were **not** re-read this pass — kept `[OPEN]`. **Test types:** AUDIT. **Gap:** 13.

ATM-067 — LLMOrchestrator AgentPool contract `[OPEN → CLOSED]`

- **Kind/Pri:** ~~AUDIT~~ ~~`[OPEN]`~~ **CLOSED (Pass 3)** · P1. **Provides:** the `AgentPool` capability-matching contract for the dev fleet. **Resolution:** read at source — **CLOSED**.
- **Verified-source `[VERIFIED-SOURCE]`:** `vasic-digital/LLMOrchestrator/pkg/agent/pool.go` — `type AgentPool interface { Acquire(ctx context.Context, requirements AgentRequirements) (Agent, error); Release(agent Agent) }`; `func NewPool() AgentPool`; siblings `multi_pool.go`, `simple_pool.go`. Capability matching (`pkg/agent/agent.go`): `Agent.Capabilities() AgentCapabilities` where `AgentCapabilities{ Vision, Streaming, ToolUse bool; MaxTokens, MaxImages int; Providers []string }` is matched against `AgentRequirements{ NeedsVision, NeedsStreaming bool; MinTokens int; PreferredAgent string }`. `pool.Acquire` blocks until an agent whose `Capabilities()` satisfy the `requirements` is free. Concrete agents: `claudecode_agent.go`, `gemini_agent.go`, `junie_agent.go`, `opencode_agent.go`, `qwencode_agent.go`. This **closes** `[OPEN: agentpool-contract]`.
- **Remaining:** adding HarmonyOS/Aurora build-agent capabilities (if those become agent tasks) is a follow-on, not a blocker. **Test types:** contract, integration. **Gap:** 2.4.

ATM-068 — FLAGGED agent-support modules verify `[OPEN – partial]`

- **Kind/Pri:** AUDIT · P1. **`[OPEN: flagged-modules-verify]`** — `Normalize`/`conversation`/`SkillRegistry`/`ToolSchema`/`MCP_Module`/`Agentic`/`Planning`/`AgentWrapper` import paths unconfirmed. **Pass-3 partial close:** `vasic-digital/Agentic` **exists** at source ("Graph-based agentic workflow orchestration", public). The remaining modules (`Normalize`, `conversation`, `SkillRegistry`, `ToolSchema`, `MCP_Module`, `Planning`, `AgentWrapper`) were **not** individually source-verified this pass — kept `[OPEN]` with the exact residual list. **Resolution:** source-verify + pin exact import paths before any item depends on them. **Test types:** AUDIT. **Gap:** 2.9, 13.

ATM-069 — Docs Chain tooling provisioning `[OPEN]`

- **Kind/Pri:** HARDEN · P1. **`[OPEN: docs-chain-tooling]`** — Docs Chain honest-SKIPs md→HTML/PDF when `pandoc` / `weasyprint` are absent (as on the current host). **Resolution:** provision pandoc/weasyprint on dev/CI hosts so `ATM-009` siblings generate. **Test types:** integration, challenges (SKIP honesty). **Gap:** 10.1.

ATM-070 — memory/HelixMemory decision (recorded)

- **Kind/Pri:** AUDIT · P2. **Decision:** Thready standardizes on `vectordb` + `embeddings` + `rag` (via the Semantic-search service ATM-039) and does **not** rely on `memory`'s word-overlap (Jaccard) search; if `memory` is later adopted, back it with `vectordb`. `HelixMemory` (Mem0/Cognee/Letta/Graphiti) is **deferred** unless full cognitive memory is required. **Gap:** 2.8.

ATM-071 — HelixStream defer (recorded)

- **Kind/Pri:** AUDIT · P2. **Decision: defer** (`HelixDevelopment/HelixStream`, SCAFFOLD, ~5 commits) — not on the Thready MVP critical path; if streaming-app testing is later required, harden the scaffold + add CI before reliance. **Gap:** 9.2.

ATM-072 — sibling-area-index cross-links `[CLOSED]`

- **Kind/Pri:** AUDIT · P1 · **CLOSED (integration pass).** The `../architecture/index.md` and `../database/index.md` canonical entry points now exist; the upstream cross-links from index.md §3 resolve. See ../INTEGRATION_REPORT.md.

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8. Estimation & sequencing guidance

The backlog is **not** sprint-timeboxed (no fixed cadence — an autonomous loop from the first prompt, `[Q6/§11.4.126]`). Instead the ruler sequences by the critical-path DAG (§2) and these rules:

- **P0 first, dependency-topological.** A track claims the highest-priority item all of whose inbound edges are `DONE`. The four foundational P0s with no product dependency — `ATM-001` (must be first), `ATM-036` (author Skills), `ATM-040` (llama embedder), `ATM-033` (OCR) — can start immediately and in parallel on separate tracks.
- **Build-new before its consumers.** `ATM-030` (callback) precedes `ATM-028` / `ATM-031` / `ATM-032`; `ATM-011` (Event Bus) precedes `ATM-025`; `ATM-040` precedes `ATM-039`; `ATM-016` precedes `ATM-017` precedes `ATM-018`. These are hard edges the ruler never inverts.
- **Spike-gated items never block the MVP claim.** `ATM-018`'s OneMe user-WS scope is `[OPEN]`; its Bot-API scope ships first so the item makes progress while the spike runs.
- **Anti-bluff gates are release-blocking, not schedule-blocking.** `ATM-052` / `ATM-058` run continuously; an item cannot reach `DONE` on a scaffold dependency without its paired-mutation gate green.

A coarse effort banding `[DEFAULT — adjustable]` for planning only (not a commitment): the six build-new services (`ATM-010/011/012/028/033/039`) are each multi-task, multi-test-type items (the “large” band); the wire/extend items (`ATM-003/004/005/019/020`) are “medium”; the audit/decision items (`ATM-070/071`) are “small”. Real sizing emerges from the RED-test-first decomposition, not up-front.

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